Learning how to prove Workshop Week 9

Groups

Group Homomorphisms

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1. Let $G_1 \times G_2$ be the direct product of two groups G_1 and G_2. We define

$$\begin{array}{lll} \pi_1 : G_1 \times G_2 \to G_1 & \text{by} & \pi_1(g_1, g_2) = g_1 \\ \iota_1 : G_1 \to G_1 \times G_2 & \text{by} & \iota(g_1) = (g_1, 1_{G_2}) \end{array}$$

Prove

(a) π_1 is an epimorphism, called the **projection onto** G_1.

(b) ι_1 is a monomorphism, called the **injection of** G_1 **onto** $G_1 \times G_2$**.**

Similarly, we can define a **projection onto** G_2, and an **injection of** G_2 **onto** $G_1 \times G_2$.

Solution: (a) Let $(g_1, g_2), (g_1', g_2') \in G_1 \times G_2$. Then

$$\pi_1\big[(g_1, g_2)(g_1', g_2')\big] = \pi(g_1g_1', g_2g_2') = g_1g_1' = \pi_1(g_1, g_2)\pi_1(g_1', g_2').$$

Thus π_1 is a homomorphism. It remains to show that π_1 is surjective but it clearly holds, since $\pi_1(g_1, 1_{G_2}) = g_1$ for all $g_1 \in G_1$.

(b) Given $g_1, g_1' \in G_1$ we have that

$$\iota(g_1g_1') = (g_1g_1', 1_{G_2}) = (g_1, 1_{G_2})(g_1', 1_{G_2}) = \iota(g_1)\iota(g_1'),$$

which shows that ι is a homomorphism. Moreover, ι is one-to-one since

$$\iota(g_1) = \iota(g_1') \Rightarrow (g_1, 1_{G_2}) = (g_1', 1_{G_2}) \Rightarrow g_1 = g_1'.$$

2. Let n be a nonzero integer, and consider the subgroup $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$ of $\mathbb{Z}$.

(a) Prove that $\mathbb{Z} \cong n\mathbb{Z}$.

(b) Show that $n\mathbb{Z} \cong m\mathbb{Z}$ for all $n \neq 0$ and $m \neq 0$.

Solution: (a) The mapping $\varphi : \mathbb{Z} \to n\mathbb{Z}$ given by $\varphi(k) = nk$ is clearly surjective, and φ is one-to-one because $\varphi(k) = \varphi(k')$ implies that $nk = nk'$, i.e., $n(k-k') = 0$ which yields that $k = k'$.

To finish, let us prove that φ is a homomorphism:

$$\varphi(k + k') = n(k + k') = nk + nk' = \varphi(k) + \varphi(k'),$$

for all $k, k' \in \mathbb{Z}$. Thus φ is an isomorphism, and so $\mathbb{Z} \cong n\mathbb{Z}$.

(b) Let $n, m \in \mathbb{Z}$ nonzero integers. By (a) we have that $\mathbb{Z} \cong n\mathbb{Z}$ and $\mathbb{Z} \cong m\mathbb{Z}$ and since $\cong$ is an equivalence relation we can conclude that $n\mathbb{Z} \cong m\mathbb{Z}$, as desired.